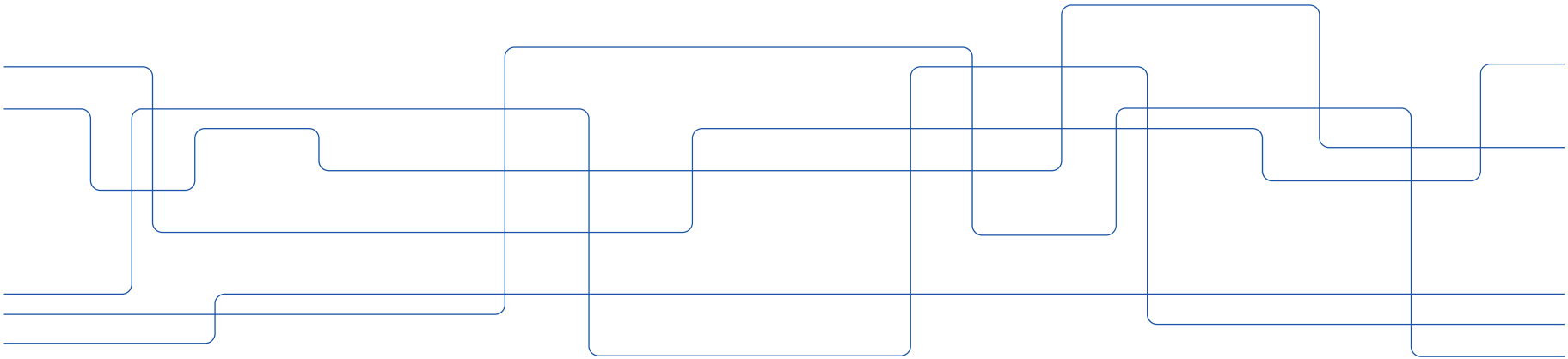
KTH ROYAL INSTITUTE OF TECHNOLOGY

CM2014: FEA Computer Laboratory

Natalia Lindgren, Neuronic Engineering, nalindgr@kth.se

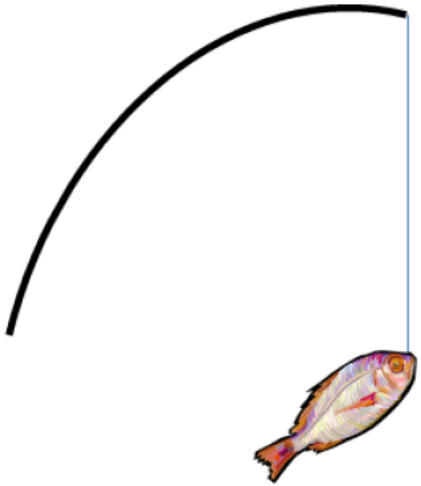


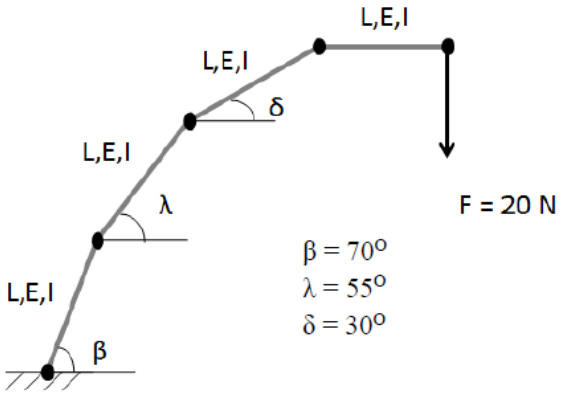
Today’s agenda

- Presentation of lab assignment
- Logistics
- Some basics and good-to-know calculation examples
- Clues and tips on how to solve the assignment
- **Questions and support!**

Assignment

Your task for this laboratory is to use your knowledge from the lectures in Finite Element Method together with Matlab to determine the deflection of a fishing rod when a fish is hooked (figure 1). Present your results with a plot of the undeformed and deformed fishing rod plus an addition table of the deflections at the nodes. You can simplify the rod with a beam. The fishing rod is 2 m with a radius of 5 cm. The cross-section of the fishing rod is circular with inertia (I) of $\frac{\pi r^4}{4}$. The Young's modulus (E) is assumed to be 100 MPa.





Logistics

- Deadline: no later than **Friday the 23rd of February**
- To **Pass** the assignment: Submit report in Canvas with correct deflections, plot and matlab code
- To **Pass with Excellence**:
  - Overall quality of report
    e.g. Language, structure, figures…
  - Matlab code quality
  - Correct results
  - ...
- **Pass with Excellence** will only we rewarded based on the first submission

dx and dz are elements of the solution (sol in MATLAB code), where dx is the $3i+1^{th}$ element and dz is the $3i+2^{th}$ element. i= 0,1,2,3.
Sol is calculated by:

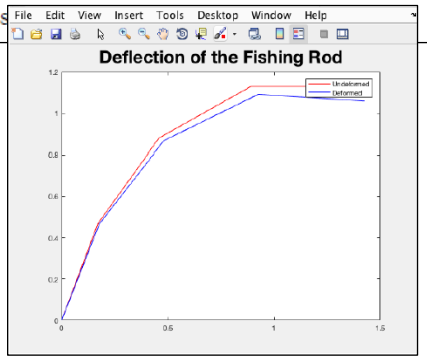
```
Sol = K_g\F;
```

W…atrix and F is…

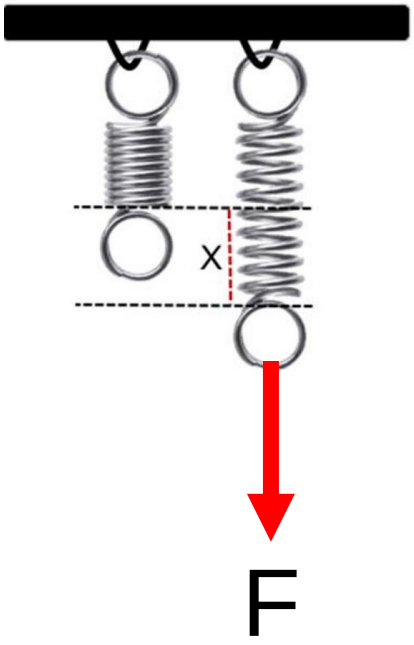
$F_e = T^T * k_e * T * D_e = K_e * D_e.$

To convert the stiffness matrices to g… culated and stored in MATLAB for … resulting in four T-matrices. The glob… calculated as

$K_{global}(n) = T^T(n) * k_e * T(n)$



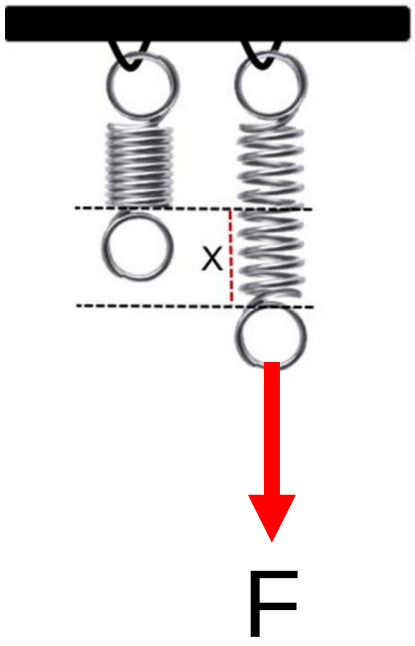
Basics: Single element



Basics: Single element

$$F = k\,x$$

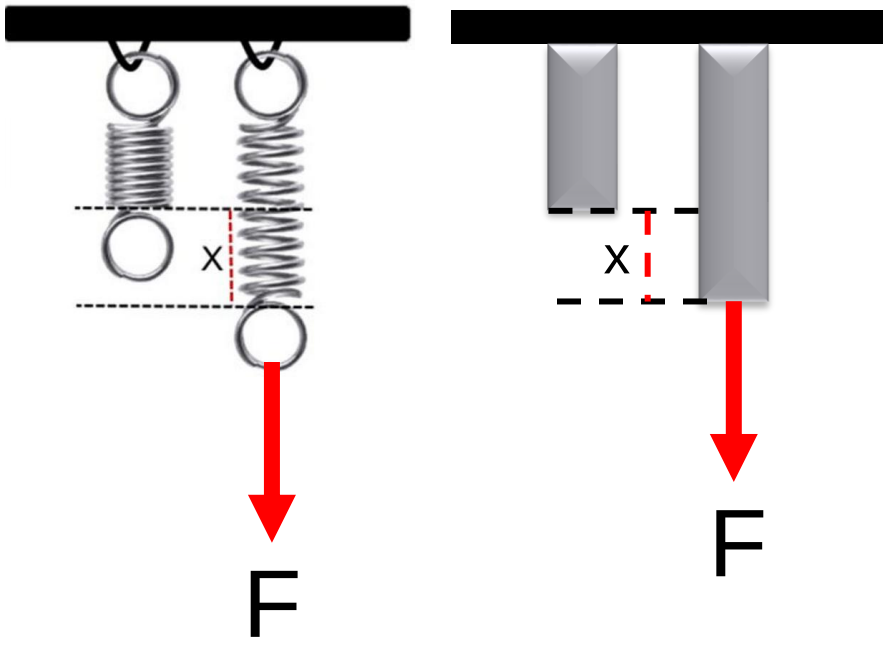
(Hooke's law)



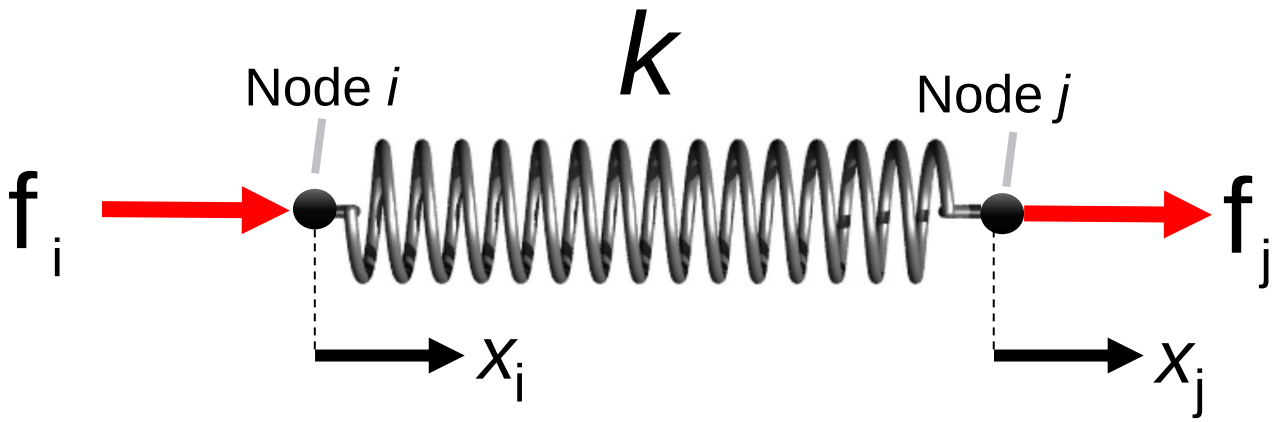
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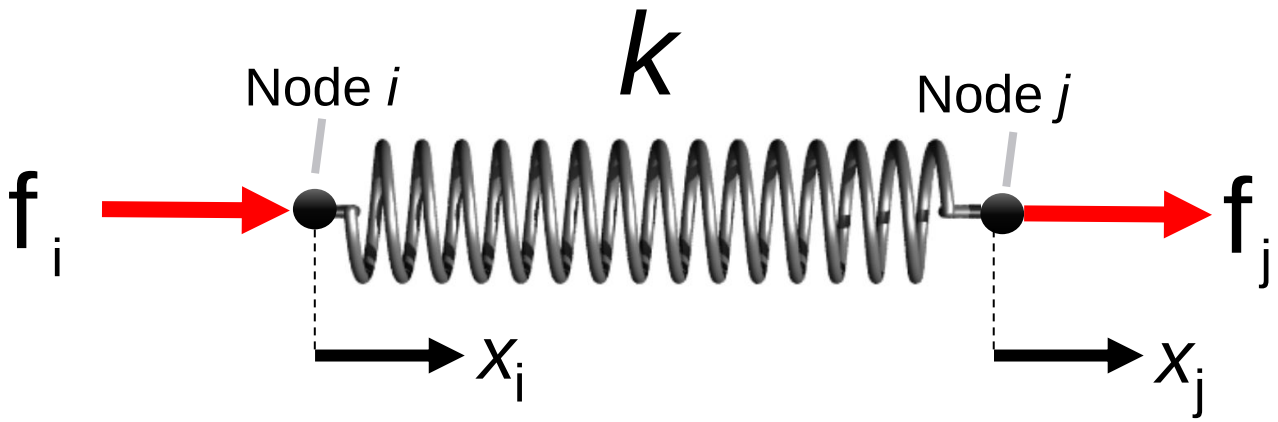
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Basics: Single element



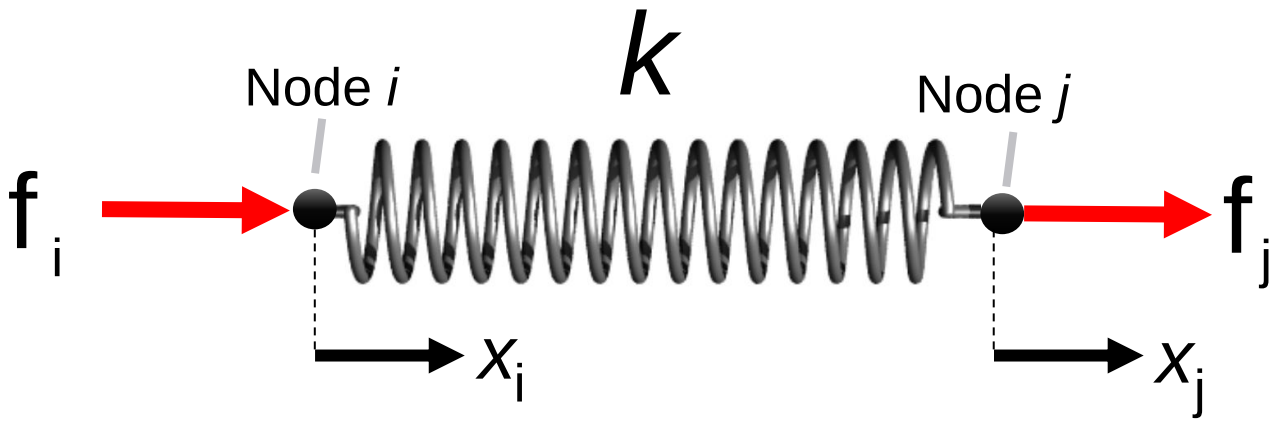
Basics: Single element



$$kx_i - kx_j = f_i$$

$$-kx_i + kx_j = f_j$$

Basics: Single element

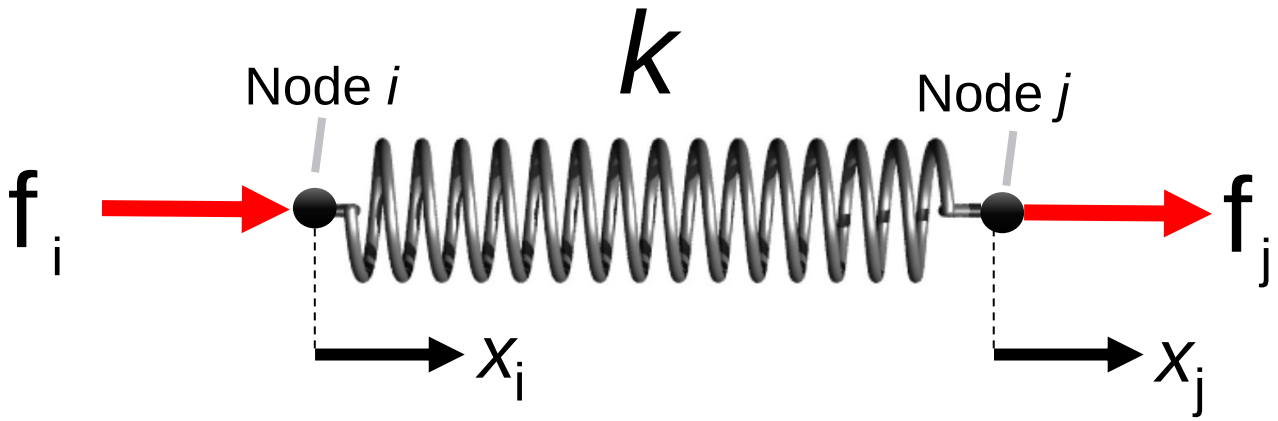


$$kx_i - kx_j = f_i$$

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$$\begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} x_i \\ x_j \end{bmatrix} = \begin{bmatrix} f_i \\ f_j \end{bmatrix}$$

Basics: Single element



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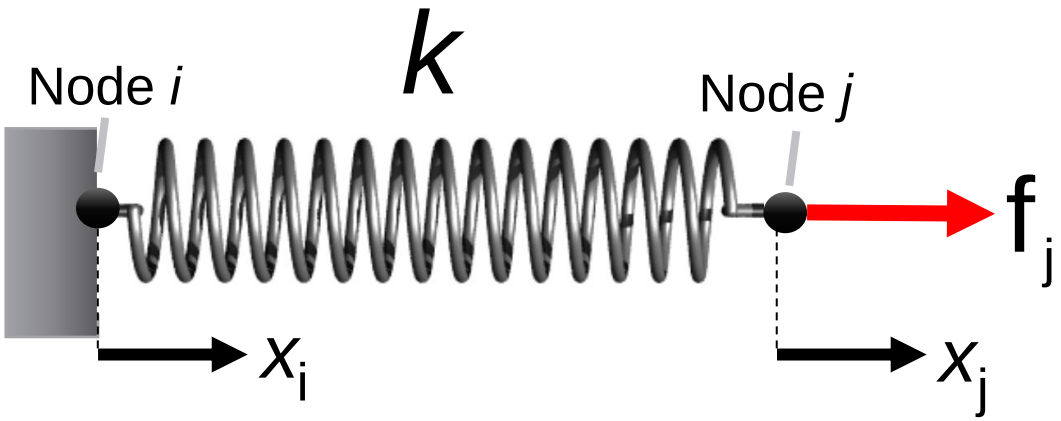
$$\begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} x_i \\ x_j \end{bmatrix} = \begin{bmatrix} f_i \\ f_j \end{bmatrix}$$

Stiffness matrix — Displacement vector

$$\mathbf{kx} = \mathbf{F}$$

Force vector

What happens if one end is fixed?



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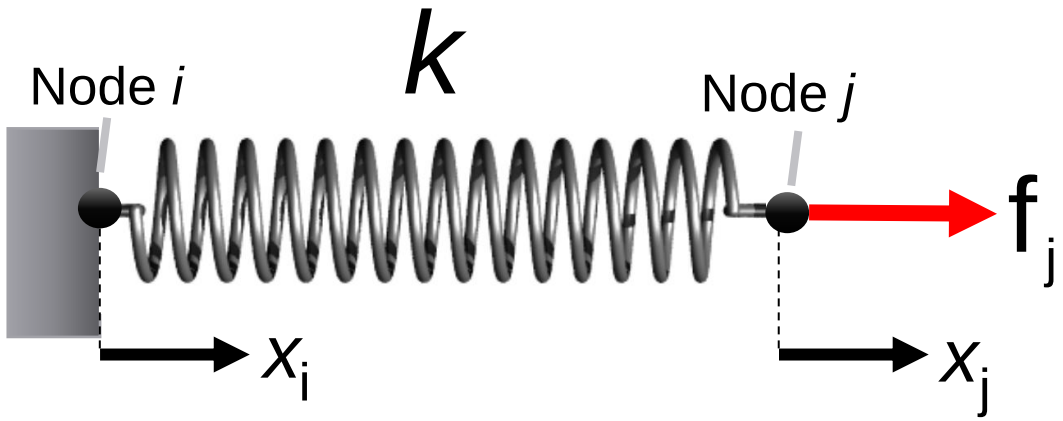
Stiffness matrix

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Stiffness matrix — Displacement vector

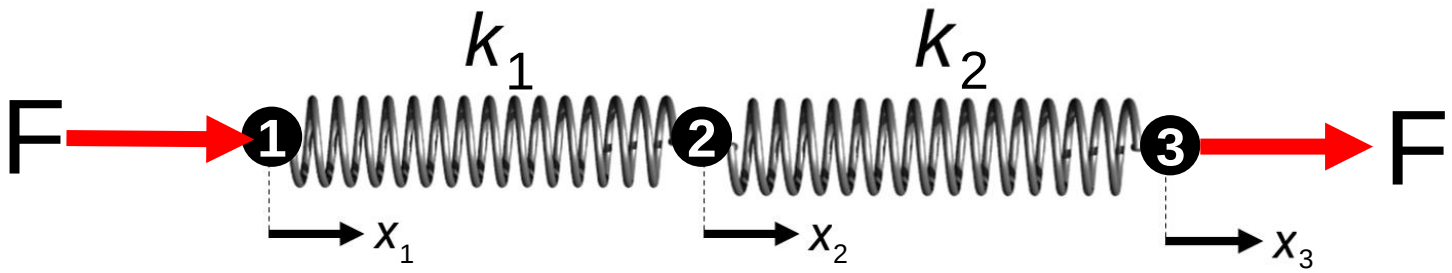
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Force vector

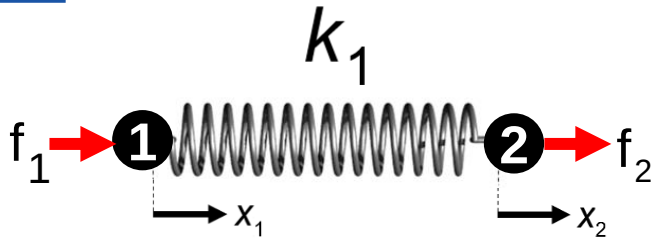
Basics: Two elements

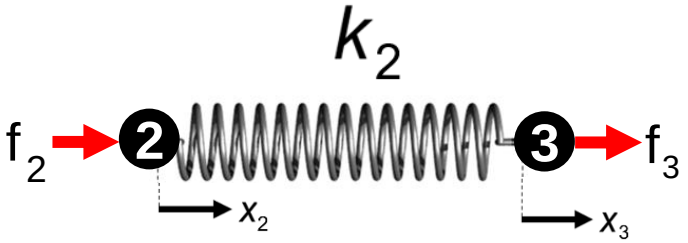


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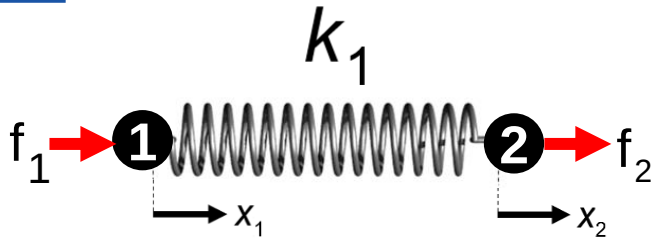


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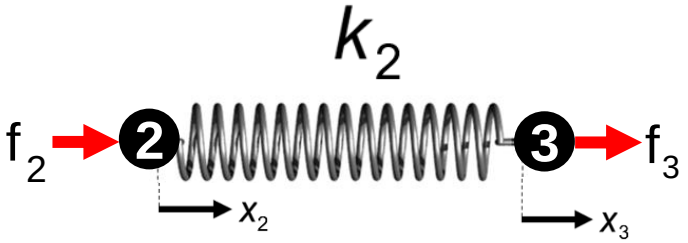




Basics: Two elements

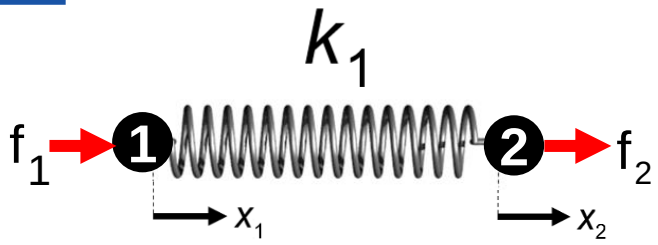


$$\begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$$



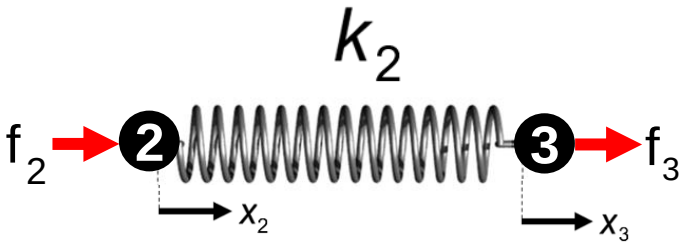
$$\begin{bmatrix} k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} f_2 \\ f_3 \end{bmatrix}$$

Basics: Two elements



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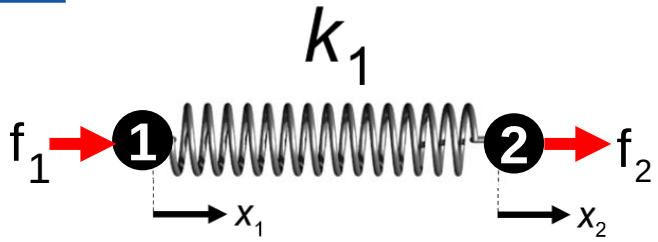
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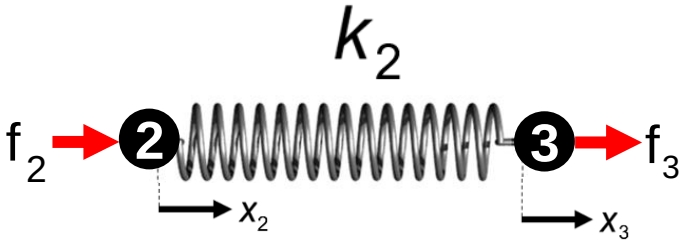


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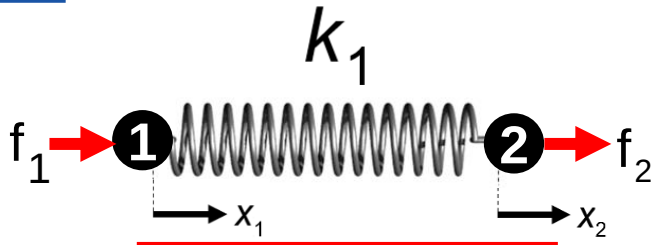
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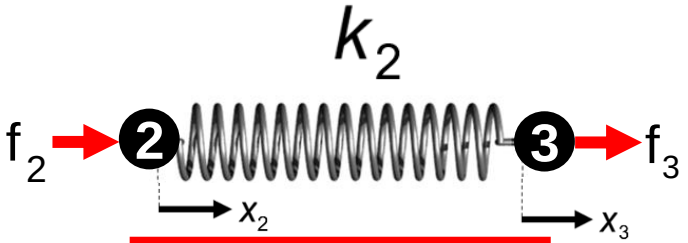
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LOCAL

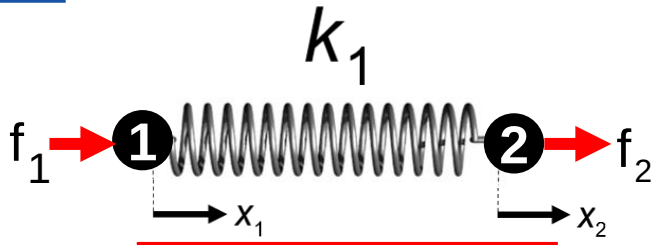


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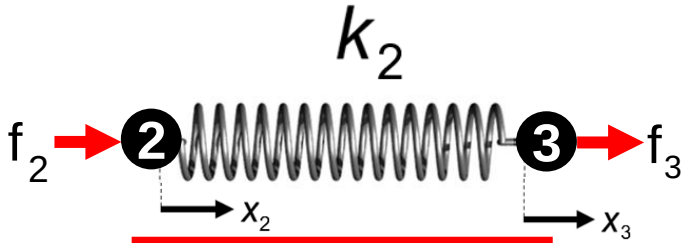
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LOCAL



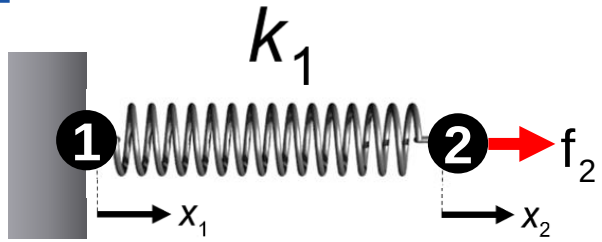
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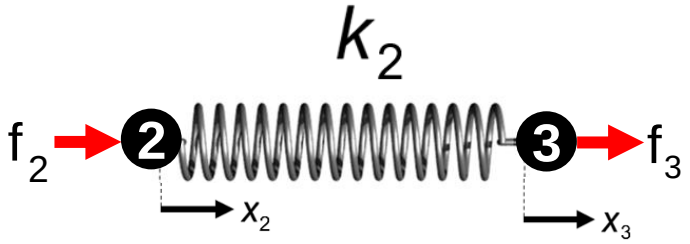
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GLOBAL

$$\begin{bmatrix} k_1 & -k_1 & 0 \\ -k_1 & k_1 + k_2 & -k_2 \\ 0 & -k_2 & k_2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

What happens if one end is fixed?





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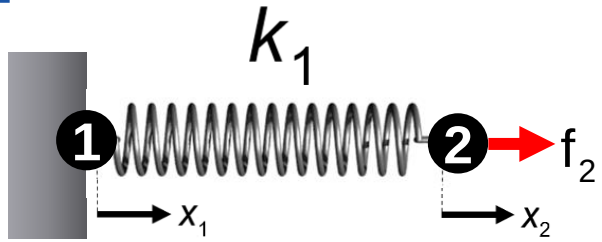
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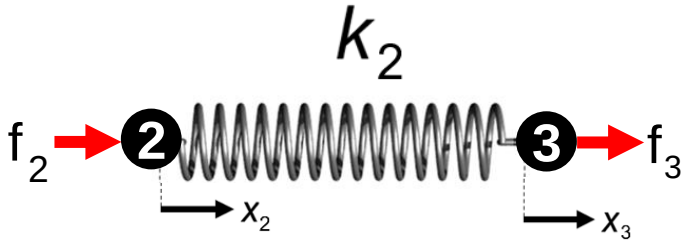
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+

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What happens if we have more DOFs?

What happens if we have more DOFs?

$$\begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$$

$$\begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12\,EI}{L^3} & \frac{6\,EI}{L^2} & 0 & -\frac{12\,EI}{L^3} & \frac{6\,EI}{L^2} \\ 0 & \frac{6\,EI}{L^2} & \frac{4\,EI}{L} & 0 & -\frac{6\,EI}{L^2} & \frac{2\,EI}{L} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12\,EI}{L^3} & -\frac{6\,EI}{L^2} & 0 & \frac{12\,EI}{L^3} & -\frac{6\,EI}{L^2} \\ 0 & \frac{6\,EI}{L^2} & \frac{2\,EI}{L} & 0 & -\frac{6\,EI}{L^2} & \frac{4\,EI}{L} \end{bmatrix} \begin{bmatrix} u_{1x} \\ w_{1z} \\ \theta_{1y} \\ u_{2x} \\ w_{2z} \\ \theta_{2y} \end{bmatrix} = \begin{bmatrix} f_{1x} \\ f_{1z} \\ M_{1y} \\ f_{2x} \\ f_{2z} \\ M_{2y} \end{bmatrix}$$

$\mathbf{k_e}$ $\mathbf{d_e}$ $\mathbf{f_e}$

What happens if we have more DOFs?

$$\begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$$

Each *k* will be a 3x3 matrix

$$\begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12\,EI}{L^3} & \frac{6\,EI}{L^2} & 0 & -\frac{12\,EI}{L^3} & \frac{6\,EI}{L^2} \\ 0 & \frac{6\,EI}{L^2} & \frac{4\,EI}{L} & 0 & -\frac{6\,EI}{L^2} & \frac{2\,EI}{L} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12\,EI}{L^3} & -\frac{6\,EI}{L^2} & 0 & \frac{12\,EI}{L^3} & -\frac{6\,EI}{L^2} \\ 0 & \frac{6\,EI}{L^2} & \frac{2\,EI}{L} & 0 & -\frac{6\,EI}{L^2} & \frac{4\,EI}{L} \end{bmatrix} \begin{bmatrix} u_{1x} \\ w_{1z} \\ \theta_{1y} \\ u_{2x} \\ w_{2z} \\ \theta_{2y} \end{bmatrix} = \begin{bmatrix} f_{1x} \\ f_{1z} \\ M_{1y} \\ f_{2x} \\ f_{2z} \\ M_{2y} \end{bmatrix}$$

$\mathbf{k_e}$ $\mathbf{d_e}$ $\mathbf{f_e}$

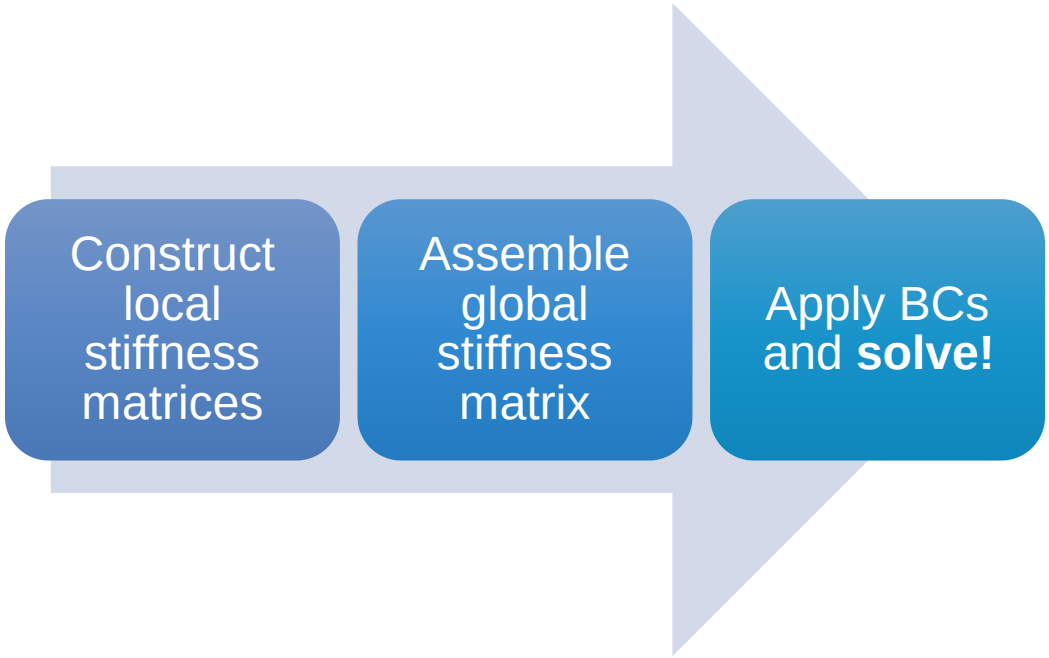
What happens if the local and global coordinate system differ?

What happens if the local and global coordinate system differ?

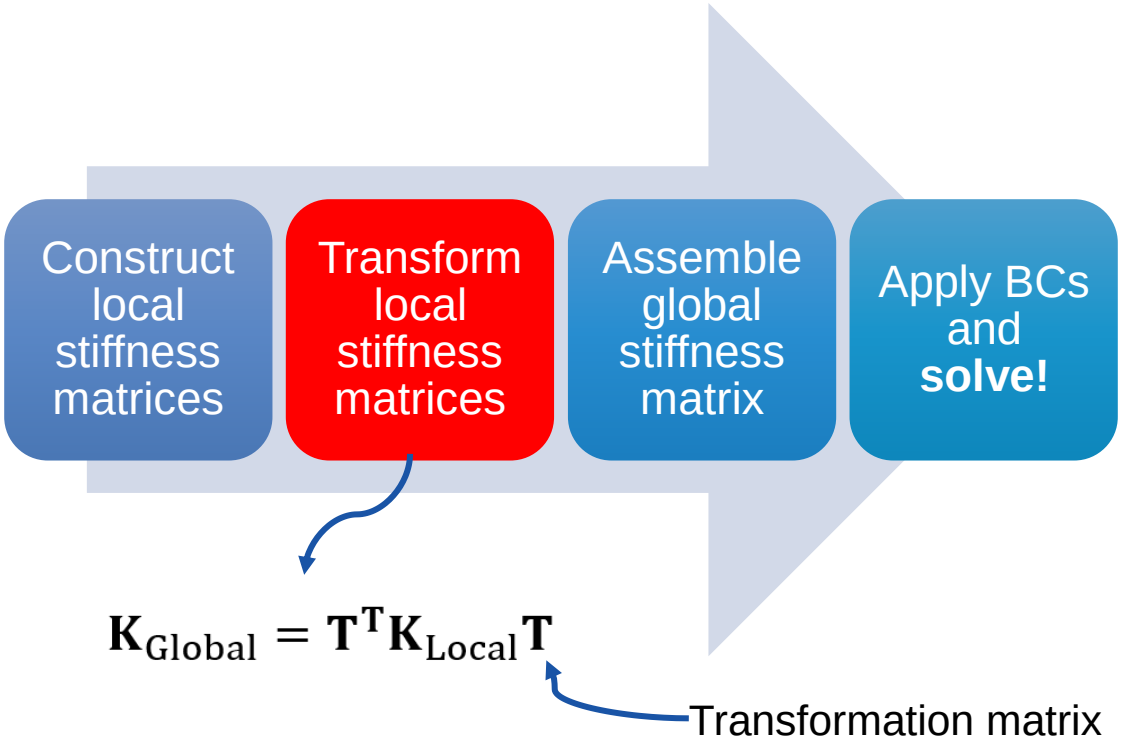
→ **Transform the local stiffness matrix!**

$$\underbrace{\begin{bmatrix} u_1 \\ w_1 \\ \theta_1 \\ u_2 \\ w_2 \\ \theta_2 \end{bmatrix}}_{\mathbf{d_e}} = \underbrace{\begin{bmatrix} \cos\alpha & \sin\alpha & 0 & 0 & 0 & 0 \\ -\sin\alpha & \cos\alpha & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos\alpha & \sin\alpha & 0 \\ 0 & 0 & 0 & -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \\ D_5 \\ D_6 \end{bmatrix}}_{\mathbf{D_e}}$$

How to solve the assignment



How to solve the assignment



Reminders

- Read the assignment uploaded to Canvas thoroughly
- Make sure to label each column and row with the number of each node
- Plug the local stiffness matrices into the global matrix according to the index number
- NOTE:
  - In your assignment each node has several **degrees of freedom**
  - The **transformation matrix** is given in the appendix
  - Example of deformation plot in the appendix

QUESTIONS?

Now...

Start with identifying the BCs, external loads and DOFs!